1st Midterm · Unit 2 · First-order qualitative analysis · 5 points · 30 min · Conceptual / fast

P02 — Logistic population with constant harvesting: equilibria against the harvest rate, the critical rate 1/4, extinction times by partial fractions and by an arctangent

A fish population $P(t)$ grows logistically with intrinsic rate $r > 0$ and carrying capacity $K > 0$ and is fished at a constant rate $H \geq 0$ (individuals per unit time), independent of the size of the stock:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right) - H.$$

With $p = P/K$, $\tau = rt$ and $h = H/(rK)$ the model becomes, writing again t for τ and $'$ for d/dt,

$$p' = p(1-p) - h, \qquad h \geq 0.$$

The model is used only while $p \geq 0$; if p reaches 0 the population is extinct (collapse). Part (c) uses $h = 2/9$ and part (d) uses $h = 1/2$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $p(t) \geq 0$ (fraction of the carrying capacity), independent variable t (dimensionless time, unit $1/r$); parameter $h \geq 0$ (dimensionless harvest).
- Operator: first order, **nonlinear** (quadratic in p), **autonomous** (f does not depend on t). $f(p) = p(1-p) - h$ is a polynomial, so f and f' are continuous and Picard's theorem gives a unique local solution for every $p(0)$; distinct solutions never cross, and equilibria split the line into invariant intervals.
- Separable: $\dfrac{dp}{f(p)} = dt$ wherever $f(p) \neq 0$.

Phase 2. Canonical formulation

$$p' = f(p), \qquad f(p) = -\left(p^2 - p + h\right) = -\left[\left(p - \frac{1}{2}\right)^2 + h - \frac{1}{4}\right].$$

The completed square already shows that the sign of $h - \frac{1}{4}$ decides everything.

Part (a) · 1 point

Equilibria and the critical harvesting rate

Find the equilibria as functions of h, the critical rate h_c and its meaning.

Step 1. Equilibria

$f(p) = 0 \iff p^2 - p + h = 0$, with discriminant $1 - 4h$:

$$p_{\pm} = \frac{1 \pm \sqrt{1 - 4h}}{2} \quad (h < \tfrac{1}{4}), \qquad p = \tfrac{1}{2} \quad (h = \tfrac{1}{4}), \qquad \text{none} \quad (h > \tfrac{1}{4}).$$

For $0 \le h < \frac{1}{4}$: $0 \le p_- < \frac{1}{2} < p_+ \le 1$, with $p_- = 0, p_+ = 1$ when $h = 0$ (the logistic equation). As h grows the two equilibria approach each other, merge at $\frac{1}{2}$ for $h = \frac{1}{4}$ and disappear.

Step 2. Critical rate and meaning

$h_c = \frac{1}{4}$. The natural growth $g(p) = p(1-p)$ has $g'(p) = 1 - 2p = 0$ at $p = \frac{1}{2}$, with $g(\frac{1}{2}) = \frac{1}{4}$, its maximum. In original variables $h = H/(rK)$, so

$$H_c = \frac{rK}{4},$$

the largest growth the stock can produce, reached at $P = K/2$. Any constant catch above it exceeds what the population can regenerate at any size (the "maximum sustainable yield"). The merging and disappearance of two equilibria as a parameter crosses a value is called a saddle-node bifurcation (extension: a name beyond the course; the course only needs the phase lines of (b)).

KEY RESULTS

- $p_{\pm} = \dfrac{1 \pm \sqrt{1 - 4h}}{2}$ for $h < \frac{1}{4}$; one equilibrium $p = \frac{1}{2}$ for $h = \frac{1}{4}$; none for $h > \frac{1}{4}$.
- $h_c = \frac{1}{4}$, i.e. $H_c = \dfrac{rK}{4}$, the maximum natural growth rate (reached at $P = K/2$).

Part (b) · 1.5 points

Phase lines in the three regimes

Describe the phase line and classify the equilibria for $h < h_c$, $h = h_c$, $h > h_c$, and say which initial populations survive.

Step 1. Case $0 \le h < \frac{1}{4}$

$f(p) = -(p - p_-)(p - p_+)$: $f < 0$ for $p < p_-$, $f > 0$ on (p_-, p_+), $f < 0$ for $p > p_+$. Phase line (arrows of p'):

$$0 \quad \longleftarrow \quad p_- \quad \longrightarrow \quad p_+ \quad \longleftarrow$$

With $f'(p) = 1 - 2p$ (Notes, p. 3, linear stability test):

$$f'(p_\pm) = 1 - \left(1 \pm \sqrt{1 - 4h}\right) = \mp\sqrt{1 - 4h},$$

so $f'(p_-) > 0$: **unstable**; $f'(p_+) < 0$: **asymptotically stable**. Populations with $p(0) > p_-$ tend to p_+; those with $p(0) < p_-$ decrease and reach 0 (p_- is a survival threshold). For $h = 0, p_- = 0$ and nothing collapses.

Step 2. Case $h = \frac{1}{4}$

$f(p) = -(p - \frac{1}{2})^2 \le 0$, with $f'(\frac{1}{2}) = 0$: the linear test is inconclusive and the sign of f decides. $p' < 0$ on both sides:

$$\longleftarrow \quad \tfrac{1}{2} \quad \longleftarrow$$

From above, p decreases to $\frac{1}{2}$ (attracting side); from below, p moves away towards 0 (repelling side). The equilibrium is **semi-stable**: it is not stable, since an arbitrarily small loss pushes the stock into collapse.

Step 3. Case $h > \frac{1}{4}$

$f(p) = -(p - \frac{1}{2})^2 - (h - \frac{1}{4}) \le -(h - \frac{1}{4}) < 0$ for every p. There is no equilibrium and every solution decreases at a rate at least $h - \frac{1}{4}$, so from any $p(0)$ it reaches 0 before the time $p(0)/(h - \frac{1}{4})$: collapse in finite time for every initial stock.

KEY RESULTS

- $h < \frac{1}{4}$: p_- unstable ($f'(p_-) = \sqrt{1 - 4h} > 0$), p_+ asymptotically stable ($f'(p_+) = -\sqrt{1 - 4h} < 0$). Survival iff $p(0) \ge p_-$ (with $p(0) = p_-$ the equilibrium itself).
- $h = \frac{1}{4}$: $p = \frac{1}{2}$ semi-stable ($f = -(p - \frac{1}{2})^2 \le 0$): attracts from above, repels below. Survival iff $p(0) \ge \frac{1}{2}$.
- $h > \frac{1}{4}$: $f \le \frac{1}{4} - h < 0$ everywhere, every solution decreases and collapses in finite time.

Part (c) · 1.5 points

Explicit solutions for h = 2/9

Solve with $p(0) = 1/2$ and with $p(0) = 1/4$; in the second case find the time of collapse.

Step 1. Factorisation and partial fractions

With $h = \frac{2}{9}$: $\sqrt{1 - \frac{8}{9}} = \frac{1}{3}$, $p_\pm = \frac{1 \pm 1/3}{2}$, i.e. $p_- = \frac{1}{3}$, $p_+ = \frac{2}{3}$, and $f(p) = -(p - \frac{1}{3})(p - \frac{2}{3})$. Write

$$\frac{1}{(p - \frac{1}{3})(p - \frac{2}{3})} = \frac{A}{p - \frac{1}{3}} + \frac{B}{p - \frac{2}{3}} \iff 1 = A\Big(p - \frac{2}{3}\Big) + B\Big(p - \frac{1}{3}\Big).$$

At $p = \frac{1}{3}$: $1 = -\frac{1}{3}A$, $A = -3$. At $p = \frac{2}{3}$: $1 = \frac{1}{3}B$, $B = 3$.

Step 2. Separation and integration

For $p \ne \frac{1}{3}, \frac{2}{3}$: $\dfrac{dp}{(p - \frac{1}{3})(p - \frac{2}{3})} = -dt$. With the antiderivatives $\int \frac{dp}{p - c} = \ln|p - c|$:

$$3 \ln \left|p - \frac{2}{3}\right| - 3 \ln \left|p - \frac{1}{3}\right| = -t + C \implies \ln \left|\frac{p - \frac{2}{3}}{p - \frac{1}{3}}\right| = -\frac{t}{3} + \frac{C}{3}.$$

Exponentiating and absorbing the sign into the constant: $\dfrac{p - \frac{2}{3}}{p - \frac{1}{3}} = K\, e^{-t/3}$, with $K = \dfrac{p(0) - \frac{2}{3}}{p(0) - \frac{1}{3}}$ from $t = 0$.

Step 3. Solving for p

Write $E = Ke^{-t/3}$: $p - \frac{2}{3} = E\,p - \frac{E}{3}$, so $p(1-E) = \frac{2}{3} - \frac{E}{3}$ and

$$p(t) = \frac{2 - K\,e^{-t/3}}{3\big(1 - K\,e^{-t/3}\big)}.$$

Step 4. The two initial data

- $p(0) = \frac{1}{2}$: $K = \dfrac{-1/6}{1/6} = -1,\ p(t) = \dfrac{2 + e^{-t/3}}{3(1 + e^{-t/3})}$. Check $p(0) = \frac{3}{6} = \frac{1}{2}$. As $t \to \infty$, $e^{-t/3} \to 0$ and $p \to \frac{2}{3} = p_+$: the stock recovers to the stable equilibrium, as the phase line predicts.
- $p(0) = \frac{1}{4}$ (below the threshold $p_- = \frac{1}{3}$): $K = \dfrac{\frac{1}{4} - \frac{2}{3}}{\frac{1}{4} - \frac{1}{3}} = \dfrac{-5/12}{-1/12} = 5$. Multiplying numerator and denominator by $e^{t/3}$:

$$p(t) = \frac{2e^{t/3} - 5}{3\big(e^{t/3} - 5\big)}.$$

Check $p(0) = \frac{-3}{-12} = \frac{1}{4}$. $p = 0$ when $2e^{t/3} = 5$:

$$T_0 = 3\ln\frac{5}{2} \approx 2.749.$$

The formula has a pole at $t = 3\ln 5 \approx 4.83 > T_0$, but the model has already stopped: the population is extinct at T_0. In original time, $t_{\text{real}} = T_0/r$.

KEY RESULTS

- Equilibria $p_- = \frac{1}{3}$ (unstable), $p_+ = \frac{2}{3}$ (stable). General solution $\dfrac{p - \frac{2}{3}}{p - \frac{1}{3}} = K\,e^{-t/3}, p = \dfrac{2 - Ke^{-t/3}}{3(1 - Ke^{-t/3})}$.
- $p(0) = \frac{1}{2}$: $K = -1, p(t) = \dfrac{2 + e^{-t/3}}{3(1 + e^{-t/3})} \to \frac{2}{3}$.
- $p(0) = \frac{1}{4}$: $K = 5, p(t) = \dfrac{2e^{t/3} - 5}{3(e^{t/3} - 5)}$, collapse at $T_0 = 3\ln\frac{5}{2} \approx 2.749$.

Part (d) · 1 point

Collapse time for h = 1/2

Take $h = 1/2$ *and* $p(0) = 1$. *Find the collapse time* T *and* $p(t)$ *on* $[0, T]$.

Step 1. Antiderivative

$p' = -\left[(p - \frac{1}{2})^2 + \frac{1}{4}\right] < 0$. With $s = 2p - 1$, $ds = 2\,dp$ and $(p - \frac{1}{2})^2 + \frac{1}{4} = \frac{s^2+1}{4}$:

$$\int \frac{dp}{(p - \frac{1}{2})^2 + \frac{1}{4}} = \int \frac{ds/2}{(s^2+1)/4} = 2\int \frac{ds}{1 + s^2} = 2\arctan s = 2\arctan(2p - 1).$$

Step 2. Time to collapse

Separating, $dt = -\dfrac{dp}{(p - \frac{1}{2})^2 + \frac{1}{4}}$. As t runs from 0 to T, p runs from 1 down to 0, so by Barrow's rule

$$T = \int_1^0 \frac{-dp}{(p - \frac{1}{2})^2 + \frac{1}{4}} = \Big[2\arctan(2p - 1)\Big]_0^1 = 2\arctan 1 - 2\arctan(-1) = 2 \cdot \frac{\pi}{4} + 2 \cdot \frac{\pi}{4} = \pi.$$

The same computation from a general $p(0)$ gives $T = 2\arctan(2p(0) - 1) - 2\arctan(-1) = 2\arctan(2p(0) - 1) + \frac{\pi}{2}$; for example $p(0) = \frac{1}{2}$ gives $\frac{\pi}{2}$.

Step 3. The solution

Integrating from 0 to t: $2\arctan(2p - 1) - 2\arctan(1) = -t$, so $\arctan(2p - 1) = \frac{\pi}{4} - \frac{t}{2}$ and

$$p(t) = \frac{1}{2} + \frac{1}{2}\tan\left(\frac{\pi}{4} - \frac{t}{2}\right), \qquad 0 \le t \le \pi.$$

Check: $p(0) = \frac{1}{2} + \frac{1}{2}\tan\frac{\pi}{4} = 1$; $p(\pi) = \frac{1}{2} + \frac{1}{2}\tan(-\frac{\pi}{4}) = 0$. By the chain rule $p' = \frac{1}{2}\sec^2(\frac{\pi}{4} - \frac{t}{2}) \cdot (-\frac{1}{2}) = -\frac{1}{4}(1 + \tan^2)$, and $-\left[(p - \frac{1}{2})^2 + \frac{1}{4}\right] = -\frac{1}{4}\tan^2 - \frac{1}{4}$: the same. In original time the collapse happens at $t_{\text{real}} = \pi/r$.

KEY RESULTS

- $T = \displaystyle\int_0^1 \frac{dp}{(p - \frac{1}{2})^2 + \frac{1}{4}} = \Big[2\arctan(2p - 1)\Big]_0^1 = \pi.$
- $p(t) = \dfrac{1}{2} + \dfrac{1}{2}\tan\left(\dfrac{\pi}{4} - \dfrac{t}{2}\right), 0 \le t \le \pi.$
- For a general start $p(0) \in [0, 1]$: $T = 2\arctan(2p(0) - 1) + \dfrac{\pi}{2}.$

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $h = 0$ recovers the logistic equation ($p_- = 0, p_+ = 1$). For $h > \frac{1}{4}$ and $p(0) = 1$, with $\omega = \sqrt{h - \frac{1}{4}}$ the same arctangent integral gives $T = \frac{2}{\omega} \arctan \frac{1}{2\omega}$ ($= \pi$ for $\omega = \frac{1}{2}$): $T \to \infty$ as $h \to \frac{1}{4}^+$ (the stock lingers near $p = \frac{1}{2}$, where p' is almost zero) and $T \to 0$ as $h \to \infty$.
- **Checks:** every explicit solution was substituted back into the equation and satisfies its initial value; the solutions agree with the phase lines of (b) (monotone, no crossing of equilibria). Dimensions: $h = H/(rK)$ is (individuals/time)/((1/time)(individuals)), dimensionless; real times are the dimensionless ones divided by r.
- **Common mistakes in the exam:** (1) applying the linear test $f'(p^*)$ at $h = \frac{1}{4}$, where it gives 0, and concluding "stable"; (2) forgetting that below p_- the population collapses even for small harvesting; (3) reading the pole of the formula of (c) as the time of extinction instead of the first zero of p; (4) losing the factor 2 from $ds = 2\,dp$ in the arctangent integral.